

HS-HU601 — Economics for Engineers

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Module 2 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Cash Flow, Interest, and Equivalence	2
1.1	Cash Flow Diagrams	2
1.2	Time Value of Money (TVM)	2
1.3	Simple vs. Compound Interest	2
1.4	Nominal and Effective Interest Rates	2
1.5	Economic Equivalence	3
2	Time Value of Money Factors	3
2.1	TVM Factors Summary Table	3
2.2	Debt Repayment (Loan Amortization)	3
3	Cash Flow and Rate of Return Analysis	4
3.1	Present Worth, Future Worth, and Annual Worth Analysis	4
3.2	Internal Rate of Return (IRR) Analysis	4
3.3	Benefit-Cost Ratio (BCR) Analysis	5
3.4	Sensitivity and Breakeven Analysis	5
4	Worked Examples	5

1. Cash Flow, Interest, and Equivalence

To evaluate engineering designs financially, we must track when money is spent and received.

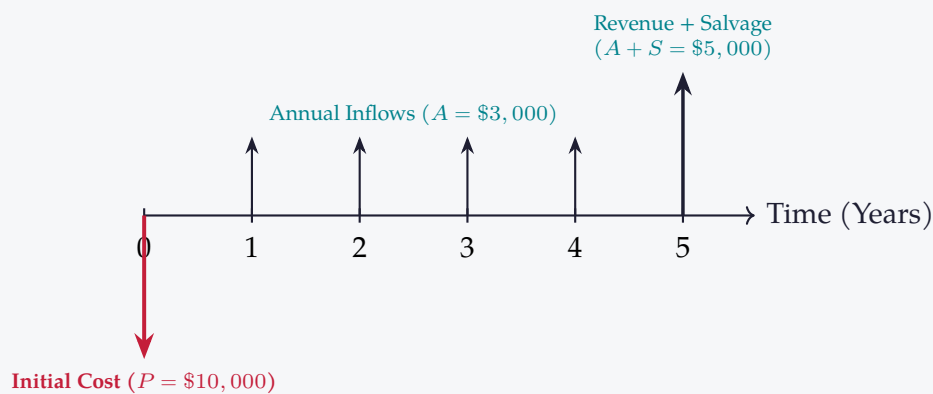
1.1 Cash Flow Diagrams

A Cash Flow Diagram (CFD) is a graphical representation of cash flows plotted on a timeline.

Cash Inflow (Receipts): Represented by vertical arrows pointing **upward** above the timeline (e.g., revenue, operating savings, salvage value).

Cash Outflow (Disbursements): Represented by vertical arrows pointing **downward** below the timeline (e.g., capital purchase, operating expenses, taxes).

Typical Cash Flow Diagram for an Engineering Project



1.2 Time Value of Money (TVM)

The **Time Value of Money** states that a dollar received today is worth more than a dollar received in the future. This is due to three factors: 1. **Opportunity Cost (Interest):** Capital can be invested to earn interest over time. 2. **Inflation:** The purchasing power of money decreases over time. 3. **Risk/Uncertainty:** Future receipts carry the risk of not being realized.

1.3 Simple vs. Compound Interest

- **Simple Interest:** Calculated **only** on the original principal amount.

$$I = P \times i \times n \quad (1)$$

- **Compound Interest:** Interest is calculated on the principal **plus** any accumulated interest from previous periods.

$$F = P \times (1 + i)^n \quad (2)$$

1.4 Nominal and Effective Interest Rates

- **Nominal Interest Rate (r):** The stated annual interest rate without considering compounding frequency within the year.

- **Effective Interest Rate (i_a):** The actual annual rate of interest paid or earned, accounting for compounding m times per year.

Effective Interest Rate Formulas

For a nominal rate r compounded m times per year:

$$i_a = \left(1 + \frac{r}{m}\right)^m - 1 \quad (3)$$

For continuous compounding ($m \rightarrow \infty$):

$$i_a = e^r - 1 \quad (4)$$

1.5 Economic Equivalence

Two cash flows (or series of cash flows) are **economically equivalent** at a given interest rate i if they have the same financial value when discounted to the same point in time.

2. Time Value of Money Factors

To simplify calculations, standardized discount factors are used to convert cash flows between the Present (P), Future (F), and Uniform Series (A).

2.1 TVM Factors Summary Table

Find / Given	Factor Name	Standard Notation	Mathematical Formula
Find F given P	Single Payment Compound Amount	$(F/P, i, n)$	$(1 + i)^n$
Find P given F	Single Payment Present Worth	$(P/F, i, n)$	$(1 + i)^{-n}$
Find F given A	Uniform Series Compound Amount	$(F/A, i, n)$	$\frac{(1 + i)^n - 1}{i}$
Find A given F	Sinking Fund	$(A/F, i, n)$	$\frac{i}{(1 + i)^n - 1}$
Find A given P	Capital Recovery	$(A/P, i, n)$	$\frac{i(1 + i)^n}{(1 + i)^n - 1}$
Find P given A	Uniform Series Present Worth	$(P/A, i, n)$	$\frac{(1 + i)^n - 1}{i(1 + i)^n}$

2.2 Debt Repayment (Loan Amortization)

Amortization is the process of paying off debt through a series of equal periodic payments. Each payment A consists of two parts: 1. **Interest Portion (I_t):** Computed on the outstanding loan balance at the beginning of period t . 2. **Principal Portion (P_t):** The

remaining part of the payment that reduces the outstanding loan balance.

Loan Amortization Equations

For a loan principal P at interest rate i repaid over n periods:

$$\text{Equal annual payment: } A = P \times (A/P, i, n) = P \times \frac{i(1+i)^n}{(1+i)^n - 1} \quad (5)$$

$$\text{Interest in period } t : I_t = B_{t-1} \times i \quad (B_{t-1} \text{ is outstanding balance}) \quad (6)$$

$$\text{Principal reduction in } t : P_t = A - I_t \quad (7)$$

$$\text{Remaining balance after } t : B_t = B_{t-1} - P_t \quad (8)$$

3. Cash Flow and Rate of Return Analysis

Engineers use standard analysis methods to determine if a project is financially viable.

3.1 Present Worth, Future Worth, and Annual Worth Analysis

- **Present Worth (PW) Analysis:** All cash inflows and outflows are discounted to the present ($t = 0$) at the Minimum Attractive Rate of Return (MARR).

$$PW(i) = \sum_{t=0}^n CF_t \times (1+i)^{-t} \quad (9)$$

Decision Rule: Accept the project if $PW(MARR) \geq 0$.

- **Future Worth (FW) Analysis:** Cash flows are compounded to a future point ($t = n$). Useful for determining net accumulation at the end of project life.
- **Annual Worth (AW) Analysis:** Cash flows are converted into an equivalent uniform series over the project life.

$$AW(i) = PW(i) \times (A/P, i, n) \quad (10)$$

Decision Rule: Accept the project if $AW(MARR) \geq 0$.

3.2 Internal Rate of Return (IRR) Analysis

The **Internal Rate of Return (IRR)** is the interest rate at which the Present Worth of all cash flows equals zero.

IRR Equation

$$PW(IRR) = \sum_{t=0}^n \frac{CF_t}{(1+IRR)^t} = 0 \quad (11)$$

Decision Rule: Accept the project if $IRR \geq MARR$.

Linear Interpolation formula: If trial rates i_1 (giving $PW_1 > 0$) and i_2 (giving $PW_2 < 0$) are close:

$$IRR \approx i_1 + \left(\frac{PW_1}{PW_1 - PW_2} \right) \times (i_2 - i_1) \quad (12)$$

3.3 Benefit-Cost Ratio (BCR) Analysis

Often used in public sector projects to assess whether the societal benefits justify the public expenditure.

Benefit-Cost Ratio Formula

$$BCR = \frac{PV(\text{Benefits})}{PV(\text{Costs})} = \frac{PW(\text{Cash Inflows})}{PW(\text{Cash Outflows})} \quad (13)$$

Decision Rule: Accept the project if $BCR \geq 1.0$.

3.4 Sensitivity and Breakeven Analysis

- **Sensitivity Analysis:** Determines how financial metrics (like PW or IRR) change when key parameters (like initial cost, useful life, or annual revenue) deviate from their estimated values.
- **Breakeven Analysis:** Determines the value of a parameter (e.g., production capacity, price, number of years) at which two alternatives become economically equivalent. At the breakeven point:

$$PW_A(i) = PW_B(i) \quad \text{or} \quad AW_A(i) = AW_B(i) \quad (14)$$

4. Worked Examples

Example 1: Nominal vs. Effective Interest

Problem: An engineering company deposits \$10,000 in a bank account that pays a nominal interest rate of 12% per year. Calculate the effective annual interest rate and the total balance after 5 years under:

- Quarterly compounding ($m = 4$).
- Continuous compounding.

Solution:

(a) Quarterly Compounding ($m = 4$):

$$i_a = \left(1 + \frac{r}{m} \right)^m - 1 = \left(1 + \frac{0.12}{4} \right)^4 - 1 = (1.03)^4 - 1$$

$$i_a \approx 1.1255 - 1 = 12.55\%$$

Future Balance after 5 years ($n = 5 \times 4 = 20$ quarters, interest rate per quarter = 3%):

$$F = P(1 + i)^{n_{\text{quarters}}} = 10,000 \times (1.03)^{20}$$

$$F \approx 10,000 \times 1.8061 = \$18,061$$

(b) Continuous Compounding:

$$i_a = e^r - 1 = e^{0.12} - 1 \approx 1.1275 - 1 = 12.75\%$$

Future Balance after 5 years:

$$F = P \times e^{r \times t} = 10,000 \times e^{0.12 \times 5} = 10,000 \times e^{0.60}$$

$$F \approx 10,000 \times 1.8221 = \$18,221$$

Answer: (a) Effective rate = **12.55%**, Balance = **\$18,061**. (b) Effective rate = **12.75%**, Balance = **\$18,221**.

Example 2: Loan Amortization Schedule

Problem: A manufacturing startup borrows \$50,000 to buy CNC tooling. The loan is to be repaid in 5 equal annual installments at a compound interest rate of 8% per year. Calculate the annual payment and find the interest and principal components for Year 1.

Solution: Identify given parameters: Principal $P = \$50,000$, Rate $i = 0.08$, Periods $n = 5$. Calculate the equal annual payment A using Capital Recovery factor ($A/P, 8\%, 5$):

$$A = P \times \frac{i(1+i)^n}{(1+i)^n - 1} = 50,000 \times \frac{0.08(1.08)^5}{(1.08)^5 - 1}$$

$$A = 50,000 \times \frac{0.08 \times 1.4693}{1.4693 - 1} = 50,000 \times 0.25046 = \$12,523$$

Calculate components for **Year 1**:

$$\text{Initial balance } B_0 = \$50,000$$

$$\text{Interest portion } I_1 = B_0 \times i = 50,000 \times 0.08 = \$4,000$$

$$\text{Principal portion } P_1 = A - I_1 = 12,523 - 4,000 = \$8,523$$

$$\text{Ending balance } B_1 = B_0 - P_1 = 50,000 - 8,523 = \$41,477$$

Answer: The equal annual payment is **\$12,523**. In Year 1, the Interest paid is **\$4,000** and the Principal reduction is **\$8,523**.

Example 3: Internal Rate of Return (IRR)

Problem: An energy efficiency project requires an initial outlay of \$25,000 and generates net annual savings of \$6,500 for 5 years. Find the IRR of this project using linear interpolation.

Solution: The cash flows are: $CF_0 = -\$25,000$, $CF_{1..5} = +\$6,500$ (Annuity). We need to find i such that $PW(i) = 0$:

$$PW(i) = -25,000 + 6,500 \times (P/A, i, 5) = 0 \quad \Rightarrow \quad (P/A, i, 5) = \frac{25,000}{6,500} \approx 3.8462$$

Let's try trial interest rates: 1. **Try** $i_1 = 9\%$:

$$(P/A, 9\%, 5) = \frac{(1.09)^5 - 1}{0.09(1.09)^5} \approx 3.8897$$

$$PW(9\%) = -25,000 + 6,500 \times 3.8897 = -25,000 + 25,283 = +\$283 \quad (\text{above zero})$$

2. **Try** $i_2 = 10\%$:

$$(P/A, 10\%, 5) = \frac{(1.10)^5 - 1}{0.10(1.10)^5} \approx 3.7908$$

$$PW(10\%) = -25,000 + 6,500 \times 3.7908 = -25,000 + 24,640 = -\$360 \quad (\text{below zero})$$

Using linear interpolation between $i_1 = 9\%$ and $i_2 = 10\%$:

$$IRR \approx i_1 + \left(\frac{PW(i_1)}{PW(i_1) - PW(i_2)} \right) \times (i_2 - i_1)$$

$$IRR \approx 0.09 + \left(\frac{283}{283 - (-360)} \right) \times (0.10 - 0.09)$$

$$IRR \approx 0.09 + \left(\frac{283}{643} \right) \times 0.01 \approx 0.09 + 0.44 \times 0.01 = 9.44\%$$

Answer: The estimated Internal Rate of Return (IRR) is **9.44%**.

Quick Revision — Module II

Interest & Rate of Return | HS-HU601

Key Formulas at a Glance

Effective Interest	$i_a = (1 + r/m)^m - 1$ (Continuous compounding: $i_a = e^r - 1$)
Single payment PW	$P = F \times (1 + i)^{-n}$
Capital Recovery	$A = P \times \frac{i(1 + i)^n}{(1 + i)^n - 1}$
Uniform series PW	$P = A \times \frac{(1 + i)^n - 1}{i(1 + i)^n}$
IRR Interpolation	$IRR \approx i_1 + \frac{PW_1}{PW_1 - PW_2} \times (i_2 - i_1)$
Benefit-Cost Ratio	$BCR = \frac{PV(\text{Benefits})}{PV(\text{Costs})} = \frac{PW(\text{Inflows})}{PW(\text{Outflows})}$

One-Line Definitions

- **Cash Flow Diagram:** Graphic representation of cash receipts (up) and disbursements (down) over time.
- **Nominal Interest Rate:** Stated annual interest rate without considering sub-year compounding.
- **Effective Interest Rate:** The actual annual rate of interest paid, accounting for compounding periods.
- **Amortization:** Paying off debt in equal installments, consisting of both principal and interest.
- **Present Worth (PW):** The value of future net cash flows discounted to the present at the MARR.
- **Annual Worth (AW):** Equivalent annual uniform series of a project's cash flows over its lifetime.
- **Internal Rate of Return:** The discount rate that makes the Present Worth of all cash flows equal zero.
- **Benefit-Cost Ratio:** Ratio of present worth of benefits to present worth of costs (must be ≥ 1.0).
- **Breakeven Point:** The value of a parameter at which two investment alternatives are economically equivalent.

Frequently Asked / Viva Questions

Q1. What is economic equivalence?

Two different cash flow streams are equivalent at a specific interest rate if they

Frequently Asked / Viva Questions (contd.)

have the same financial value when discounted to the same point in time.

Q2. What is the difference between nominal and effective interest rates?

Nominal rate is the annual rate ignoring sub-year compounding frequency. Effective rate represents the actual annual interest earned, accounting for compounding within the year.

Q3. How does the principal and interest portion change in loan amortization?

Over time, as the outstanding principal balance decreases, the interest portion of the fixed payment decreases, while the principal repayment portion increases.

Q4. Define the Sinking Fund factor ($A/F, i, n$).

It determines the uniform annual payment (A) needed to accumulate a specific future sum (F) after n years at interest rate i .

Q5. State the decision criteria for Present Worth and Annual Worth analysis.

A project is economically viable if its Present Worth (PW) or Annual Worth (AW), evaluated at the MARR, is greater than or equal to zero (≥ 0).

Q6. What is the Minimum Attractive Rate of Return (MARR)?

The minimum interest rate a company requires on its capital investments before accepting a project, based on its cost of capital and risk.

Q7. How is the Internal Rate of Return (IRR) calculated?

By setting the $PW(i) = 0$ and solving for i . In practice, it is solved by finding two rates where PW changes sign and interpolating linearly between them.

Q8. Why is continuous compounding relevant in engineering economics?

It represents the limit as compounding frequency approaches infinity. It is used in advanced financial modeling and for automated, continuous transactions.

Q9. Explain the Benefit-Cost Ratio decision rule.

If $BCR \geq 1.0$, the discounted benefits exceed the discounted costs, meaning the project is acceptable. If $BCR < 1.0$, the project is rejected.

Q10. What is the purpose of sensitivity analysis in engineering designs?

It assesses how variations or uncertainty in estimated parameters (e.g., initial cost, salvage value, inflation) affect the project's profitability.

Q11. Define the Capital Recovery factor ($A/P, i, n$) and explain its economic meaning.

It represents the equivalent annual cost of owning an asset, combining both the initial purchase depreciation and the interest cost on the invested capital.

Q12. What occurs at the breakeven point between two engineering alternatives?

The two alternatives become economically equivalent, meaning their present worths or annual worths are identical. Choosing either alternative yields the same economic result.

Comparison of TVM Factors and Operations

Factor Symbol	Name	Core Formula	Typical Application
$(F/P, i, n)$	Compound Amount	$(1 + i)^n$	Finding the future value of a single deposit.
$(P/F, i, n)$	Present Worth	$(1 + i)^{-n}$	Discounting a single future payment back to present.
$(F/A, i, n)$	Uniform Compound	$\frac{(1 + i)^n - 1}{i}$	Accumulating a pension fund through yearly deposits.
$(A/P, i, n)$	Capital Recovery	$\frac{i(1 + i)^n}{(1 + i)^n - 1}$	Determining equal annual mortgage or loan payments.
$(P/A, i, n)$	Uniform Present	$\frac{(1 + i)^n - 1}{i(1 + i)^n}$	Evaluating the present worth of uniform annual savings.